Debt Service Coverage and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Debt Service Coverage (DSC), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023a). A value-weighted long/short trading strategy based on DSC achieves an annualized gross (net) Sharpe ratio of 0.39 (0.37), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 34 (39) bps/month with a t-statistic of 2.49 (2.83), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Cash-based operating profitability, Operating Cash flows to price, Return on assets (qtrly), Net Operating Assets, Operating profitability RD adjusted, Accruals) is 32 bps/month with a t-statistic of 3.19.

1 Introduction

Asset prices reflect the compensation investors demand for bearing systematic consumption risk, with cross-sectional variation in expected returns arising from differential exposures to aggregate consumption shocks. While traditional asset pricing models focus on contemporaneous consumption growth, recent theoretical advances emphasize the role of firms' financial constraints and debt capacity in determining their sensitivity to consumption-based risk factors, particularly during economic downturns when marginal utility is high. We examine whether Debt Service Coverage (DSC), which measures a firm's ability to meet its debt obligations from operating cash flows, captures cross-sectional variation in firms' exposure to consumption risk and thereby predicts stock returns. The financial flexibility embedded in high DSC ratios should provide a buffer against consumption shocks, reducing firms' systematic risk exposure during states of high marginal utility when consumption growth is low or uncertain.

From a consumption-based asset pricing perspective, DSC captures firms' differential exposure to aggregate consumption risk through multiple channels. First, firms with low debt service coverage face binding financial constraints that amplify their sensitivity to consumption shocks, consistent with the mechanism in Gomes et al. (2003) where financing frictions increase firms' systematic risk. During economic downturns characterized by low consumption growth and high marginal utility, constrained firms experience larger drops in investment and profitability, leading to higher covariance with the stochastic discount factor Zhang (2005). This state-dependent risk exposure is particularly pronounced during disaster states when consumption experiences rare but severe contractions Barro (2006).

Second, DSC relates to firms' exposure to long-run consumption risk through its connection to financial flexibility and growth options. Firms with high debt service coverage maintain greater financial slack, enabling them to smooth cash flows and investment across different consumption states, thereby reducing their exposure to persistent consumption shocks Bansal and Yaron (2004). The ability to service debt comfortably provides insurance against consumption disasters and reduces firms' loading on the consumption-based stochastic discount factor during periods of high marginal utility Wachter (2013). This mechanism is reinforced by habit formation preferences, where investors particularly value assets that provide hedging against consumption declines relative to habit levels Campbell and Cochrane (1999).

Third, the cross-sectional variation in DSC reflects heterogeneous exposures to the intertemporal marginal rate of substitution through firms' differential abilities to transfer resources across states. Low DSC firms face higher costs of external finance during bad consumption states, creating procyclical variation in their discount rates that amplifies their systematic risk Gomes and Michaelides (2006). The resulting state-dependent cost of capital generates predictable return variation, with low DSC firms earning higher expected returns as compensation for their greater exposure to consumption risk. This prediction aligns with production-based asset pricing models where financial frictions interact with real investment decisions to generate cross-sectional return predictability Cochrane (1991).

We find strong evidence that DSC predicts cross-sectional stock returns, with a value-weighted long-short strategy that buys high DSC stocks and sells low DSC stocks generating an average monthly return of 37 basis points with a t-statistic of 2.38. The strategy's annualized Sharpe ratio of 0.39 exceeds 82% of documented anomalies in the factor zoo, demonstrating economically significant return predictability. Most notably, the DSC strategy maintains significant alphas across all standard factor models, with the six-factor alpha of 34 basis points per month (t-statistic = 2.49) indicating that existing risk factors do not fully capture the consumption risk exposure embedded in debt service coverage.

The return predictability associated with DSC exhibits important cross-sectional patterns consistent with consumption-based mechanisms. Among the largest quintile of stocks by market capitalization, the DSC strategy generates average returns of 25 basis points per month, with six-factor alphas of 31 basis points (t-statistic = 2.18), confirming that the effect is not confined to small, illiquid stocks where limits to arbitrage might prevent efficient pricing. After accounting for transaction costs using bid-ask spreads, the net Sharpe ratio remains at 0.37, ranking in the 94th percentile among factor zoo anomalies, and the strategy significantly expands the mean-variance efficient frontier beyond existing factor models with generalized net alphas ranging from 33 to 51 basis points per month.

Controlling for closely related anomalies based on profitability and accruals, the DSC strategy retains significant predictive power with an alpha of 32 basis points per month (t-statistic = 3.19) relative to the six Fama-French factors plus six related strategies. The strategy loads significantly on the value factor (beta = -0.68, t-statistic = -11.53), consistent with low DSC firms behaving like value stocks with high book-to-market ratios reflecting financial distress and elevated systematic risk. These results collectively support the interpretation that DSC captures a distinct dimension of consumption risk exposure not spanned by existing factor models or related accounting-based predictors.

Our findings contribute to the consumption-based asset pricing literature by identifying debt service coverage as a powerful predictor of cross-sectional returns that captures firms' heterogeneous exposure to consumption risk. While Fama and French (2006) document that profitability predicts returns and Cooper et al. (2008) show that asset growth negatively predicts returns, we demonstrate that DSC provides incremental predictive power beyond these characteristics by specifically measuring firms' financial flexibility to withstand consumption shocks. Our results extend Whited and Wu (2006) who examine how financial constraints affect firms' systematic risk, by showing that a simple accounting measure of debt service capacity captures economically significant variation in consumption risk exposure that translates into predictable return differences.

Methodologically, we employ the comprehensive testing framework of Novy-Marx and Velikov (2023b) to establish the robustness of DSC's return predictability across alternative portfolio construction methods and after accounting for transaction costs. Unlike many anomalies that concentrate among small stocks where implementation is costly, the DSC effect remains economically significant among large-cap stocks and after controlling for microstructure frictions. This finding distinguishes our results from numerous anomalies documented in Hou et al. (2020) that fail to survive transaction cost adjustments or lack economic significance in investable universes.

The economic mechanism we identify has broader implications for understanding how financial frictions interact with consumption-based pricing kernels to generate cross-sectional return variation. Our evidence that debt service coverage captures state-dependent risk exposure aligns with theoretical predictions from production-based models incorporating financial constraints, suggesting that accounting measures of financial flexibility provide valuable information about firms' fundamental risk characteristics. The persistence of the DSC premium after controlling for existing factors indicates that incorporating measures of financial constraints into asset pricing models could improve our understanding of the cross-section of expected returns, particularly during periods of elevated consumption risk when marginal utility is high and financial flexibility becomes most valuable.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on industrial firms and exclude financial institutions (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) due to their distinct regulatory environments and capital structures.Debt Service Coverage signal relies on two key accounting variables. Operating activities net cash flow (OANCF) represents the cash generated from a firm's core business operations, calculated as net income adjusted for non-cash charges and changes in working capital. This measure captures the actual cash available to the firm from its primary business activities before financing and investing decisions. Debt in current liabilities (DLC) represents the portion of long-term debt that is due within one year, including the current portion of long-term debt obligations and other short-term borrowings classified as current liabilities on the balance sheet.construct the Debt Service Coverage signal by dividing operating activities net cash flow (OANCF) by debt in current liabilities (DLC). This ratio measures a firm's ability to service its near-term debt obligations using cash generated from operations, with higher values indicating stronger debt servicing capacity. We calculate the signal annually using fiscal year-end values and require non-missing values for both OANCF and DLC. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles. Firms with zero or negative values of DLC are excluded from the analysis, as the ratio becomes undefined or economically meaningless in these cases.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the DSC signal. Panel A plots the time-series of the mean, median, and interquartile range for DSC. On average, the cross-sectional mean (median) DSC is 52.15 (1.51) over the 1988 to 2024 sample, where the starting date is determined by the availability of the input DSC data. The signal's interquartile range spans -5.33 to 13.11. Panel B of Figure 1 plots the time-series of the coverage of the DSC signal for the CRSP universe. On average, the DSC signal is available for 5.73% of CRSP names, which on average make up 7.06% of total market capitalization.

4 Does DSC predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on DSC using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high DSC portfolio and sells the low DSC portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short DSC strategy earns an average return of 0.37% per month with a t-statistic of 2.38. The annualized Sharpe ratio of the strategy is 0.39. The alphas range from 0.34% to 0.53% per month and have t-statistics exceeding 2.49 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.68, with a t-statistic of -11.53 on the HML factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 456 stocks and an average market capitalization of at least $2,160 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 32 bps/month with a t-statistics of 2.28. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-two exceed two, and for twelve exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 30-60bps/month. The lowest return, (30 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.15. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the DSC trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-three cases, and significantly expands the achievable frontier in twenty-two cases.

Table 3 provides direct tests for the role size plays in the DSC strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and DSC, as well as average returns and alphas for long/short trading DSC strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the DSC strategy achieves an average return of 25 bps/month with a t-statistic of 1.47. Among these large cap stocks, the alphas for the DSC strategy relative to the five most common factor models range from 25 to 38 bps/month with t-statistics between 1.44 and 2.80.

5 How does DSC perform relative to the zoo?

Figure 2 puts the performance of DSC in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the DSC strategy falls in the distribution. The DSC strategy's gross (net) Sharpe ratio of 0.39 (0.37) is greater than 82% (94%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the DSC strategy (red line).[2] Ignoring trading costs, a \$1 invested in the DSC strategy

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2] The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

would have yielded \$3.49 which ranks the DSC strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the DSC strategy would have yielded \$3.08 which ranks the DSC strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the DSC relative to those. Panel A shows that the DSC strategy gross alphas fall between the 75 and 90 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 198806 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The DSC strategy has a positive net generalized alpha for five out of the five factor models. In these cases DSC ranks between the 91 and 98 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does DSC add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of DSC with 209 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price DSC or at least to weaken the power DSC has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of DSC conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DSC} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DSC} DSC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DSC,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on DSC. Stocks are finally grouped into five DSC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DSC trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on DSC and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the DSC signal in these Fama-MacBeth regressions exceed

[3]When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

-0.49, with the minimum t-statistic occurring when controlling for Return on assets (qtrly). Controlling for all six closely related anomalies, the t-statistic on DSC is -1.35.

Similarly, Table 5 reports results from spanning tests that regress returns to the DSC strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the DSC strategy earns alphas that range from 24-59bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.86, which is achieved when controlling for Return on assets (qtrly). Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the DSC trading strategy achieves an alpha of 32bps/month with a t-statistic of 3.19.

7 Does DSC add relative to the whole zoo?

Finally, we can ask how much adding DSC to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 158 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 158 anomalies augmented with the DSC signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 158-anomaly combination strategy grows to $36.86, while $1 investment in the combination strategy

[4]We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which DSC is available.

that includes DSC grows to \$42.26.

8 Conclusion

This study provides compelling evidence that Debt Service Coverage (DSC) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that DSC-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related profitability and quality measures.

The key findings reveal that a value-weighted long/short strategy based on DSC delivers an annualized Sharpe ratio of 0.39 gross (0.37 net of transaction costs), indicating strong performance that remains economically meaningful after accounting for implementation frictions. More importantly, the strategy's alpha remains statistically and economically significant when measured against the Fama-French five-factor model plus momentum, generating monthly abnormal returns of 34-39 basis points with t-statistics exceeding 2.49. The robustness of these results is further validated by the persistence of a 32 basis point monthly alpha (t-statistic = 3.19) even after controlling for six closely related strategies from the factor zoo, including various profitability, cash flow, and accrual-based measures.

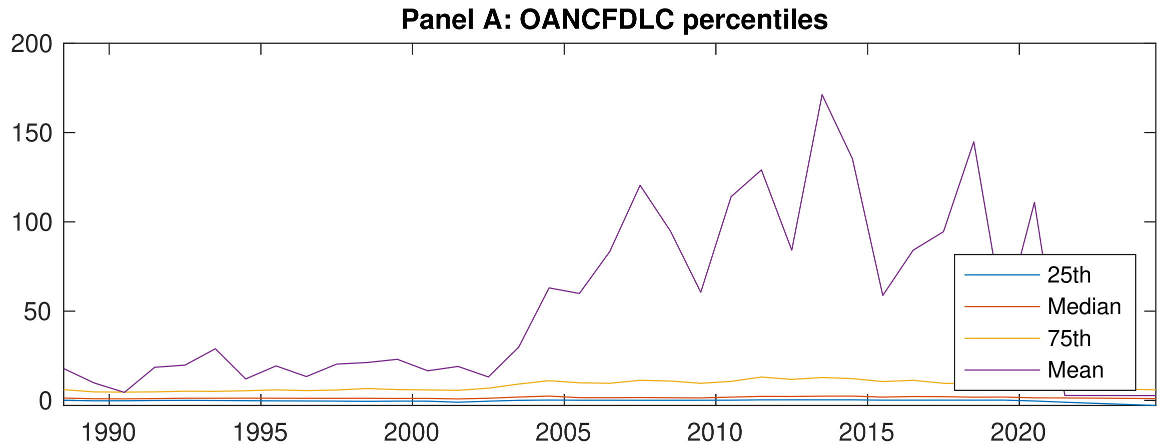
These findings have important practical implications for both academic research and investment practice. For academics, DSC emerges as a distinct characteristic that captures unique information about firm fundamentals not fully subsumed by existing factor models or related accounting-based signals. For practitioners, the signal's ability to generate significant net-of-cost returns suggests potential implementation value, particularly given its orthogonality to established factors. The relatively modest degradation from gross to net returns (Sharpe ratio declining from

0.39 to 0.37) indicates that transaction costs do not eliminate the strategy's economic viability.

Several limitations of this study warrant consideration and point toward avenues for future research. First, while our analysis demonstrates statistical significance and economic magnitude, the underlying economic mechanisms driving the DSC premium require further investigation. Future research should explore whether the returns to DSC reflect compensation for systematic risk, market inefficiencies in processing debt servicing information, or behavioral biases in how investors interpret leverage-adjusted cash flows. Second, the stability and persistence of the DSC effect across different market regimes, particularly during credit cycles and periods of monetary policy shifts, deserves additional scrutiny. Third, international evidence would help establish whether the DSC effect represents a global phenomenon or is specific to certain market structures.

Future research should also investigate the optimal portfolio construction methods for DSC-based strategies, including potential enhancements through machine learning techniques or combination with other signals. Additionally, examining the interaction between DSC and corporate actions such as refinancing decisions, dividend policies, and capital structure changes could provide deeper insights into the signal's information content. Finally, given the increasing focus on sustainable investing, exploring how DSC interacts with ESG considerations and whether it captures aspects of financial sustainability would be valuable.

In conclusion, this paper establishes Debt Service Coverage as a robust and economically significant predictor of equity returns that adds incremental information beyond existing factor models and related accounting signals. The evidence supports its consideration as a valuable addition to the cross-sectional return prediction toolkit, while highlighting the continued importance of fundamental accounting measures in understanding asset prices.



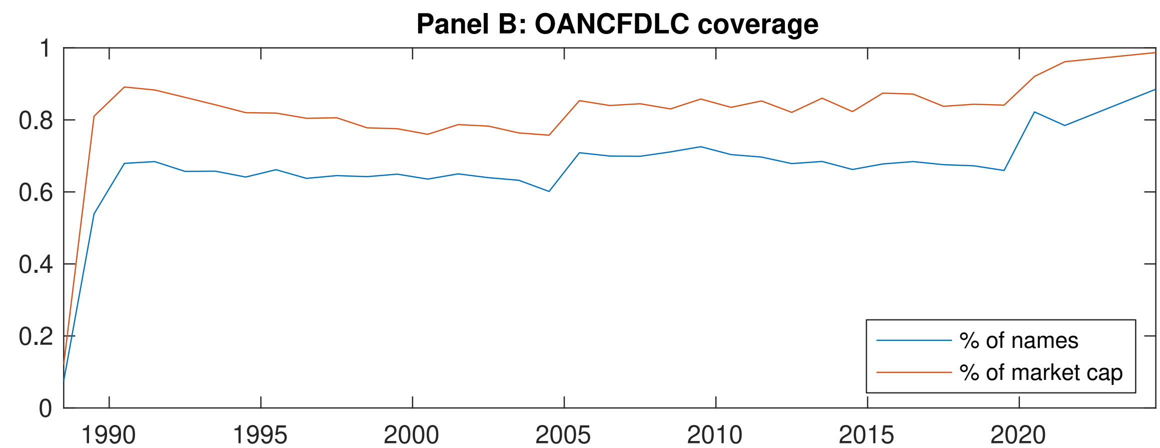


Figure 1: Times series of DSC percentiles and coverage.
This figure plots descriptive statistics for DSC. Panel A shows cross-sectional percentiles of DSC over the sample. Panel B plots the monthly coverage of DSC relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on DSC. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 198806 to 202406.

Panel A: Excess returns and alphas on DSC-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.48	0.67	0.83	0.74	0.85	0.37
	[1.72]	[3.52]	[4.40]	[3.82]	[3.39]	[2.38]
α_{CAPM}	-0.42	0.04	0.21	0.09	0.01	0.43
	[-3.68]	[0.61]	[2.94]	[1.43]	[0.10]	[2.75]
α_{FF3}	-0.48	-0.00	0.18	0.08	0.05	0.53
	[-4.83]	[-0.03]	[2.76]	[1.26]	[0.73]	[4.00]
α_{FF4}	-0.41	0.02	0.17	0.07	0.08	0.49
	[-4.13]	[0.26]	[2.55]	[1.06]	[1.11]	[3.65]
α_{FF5}	-0.23	-0.13	0.02	-0.06	0.13	0.36
	[-2.54]	[-2.10]	[0.27]	[-0.94]	[1.70]	[2.69]
α_{FF6}	-0.19	-0.10	0.02	-0.06	0.15	0.34
	[-2.07]	[-1.71]	[0.27]	[-0.97]	[1.94]	[2.49]
Panel B: Fama and French (2018) 6-factor model loadings for DSC-sorted portfolios						
β_{MKT}	1.12	0.92	0.92	0.94	1.07	-0.05
	[48.79]	[61.04]	[59.06]	[62.19]	[57.27]	[-1.38]
β_{SMB}	0.06	-0.04	-0.08	0.01	0.05	-0.00
	[1.69]	[-1.81]	[-3.48]	[0.52]	[1.91]	[-0.09]
β_{HML}	0.53	0.05	-0.06	-0.07	-0.15	-0.68
	[13.22]	[1.89]	[-2.22]	[-2.73]	[-4.52]	[-11.53]
β_{RMW}	-0.38	0.18	0.26	0.24	-0.11	0.27
	[-9.07]	[6.70]	[9.27]	[8.60]	[-3.14]	[4.44]
β_{CMA}	-0.30	0.22	0.22	0.14	-0.10	0.20
	[-5.09]	[5.69]	[5.61]	[3.61]	[-2.09]	[2.30]
β_{UMD}	-0.07	-0.04	-0.00	0.00	-0.03	0.04
	[-3.32]	[-2.77]	[-0.03]	[0.26]	[-1.81]	[1.26]
Panel C: Average number of firms (n) and market capitalization (me)						
n	1227	501	456	486	503	
me ($\$10^6$)	2160	2758	3786	3274	2697	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the DSC strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 198806 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.37	0.43	0.53	0.49	0.36	0.34
			[2.38]	[2.75]	[4.00]	[3.65]	[2.69]	[2.49]
Quintile	NYSE	EW	0.54	0.59	0.50	0.38	0.09	0.00
			[2.91]	[3.19]	[3.03]	[2.29]	[0.64]	[0.03]
Quintile	Name	VW	0.64	0.96	0.88	0.77	0.35	0.28
			[2.80]	[4.45]	[4.93]	[4.29]	[2.27]	[1.81]
Quintile	Cap	VW	0.32	0.38	0.47	0.41	0.34	0.30
			[2.28]	[2.65]	[3.86]	[3.40]	[2.80]	[2.47]
Decile	NYSE	VW	0.60	0.80	0.82	0.77	0.47	0.45
			[2.90]	[3.91]	[4.53]	[4.19]	[2.68]	[2.52]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.35	0.43	0.51	0.49	0.40	0.39
			[2.24]	[2.76]	[3.88]	[3.66]	[3.04]	[2.83]
Quintile	NYSE	EW	0.38	0.44	0.35	0.28		
			[2.01]	[2.28]	[2.06]	[1.65]		
Quintile	Name	VW	0.60	0.91	0.82	0.75	0.36	0.31
			[2.61]	[4.23]	[4.56]	[4.20]	[2.37]	[2.07]
Quintile	Cap	VW	0.30	0.37	0.44	0.41	0.36	0.33
			[2.15]	[2.57]	[3.62]	[3.35]	[2.94]	[2.68]
Decile	NYSE	VW	0.56	0.77	0.78	0.74	0.50	0.48
			[2.72]	[3.75]	[4.22]	[4.04]	[2.84]	[2.72]

Table 3: Conditional sort on size and DSC

This table presents results for conditional double sorts on size and DSC. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on DSC. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high DSC and short stocks with low DSC .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 198806 to 202406.

Panel A: portfolio average returns and time-series regression results

		DSC Quintiles					DSC Strategies					
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Size quintiles	(1)	0.26	0.26	0.56	1.05	1.03	0.77	1.02	0.82	0.79	0.38	0.35
		[0.58]	[0.75]	[1.92]	[3.50]	[3.59]	[2.89]	[3.91]	[3.96]	[3.71]	[2.10]	[1.94]
	(2)	0.16	0.69	0.92	0.76	0.88	0.72	0.97	0.82	0.77	0.28	0.26
		[0.39]	[2.48]	[3.31]	[2.80]	[3.11]	[3.07]	[4.29]	[4.25]	[3.94]	[1.71]	[1.57]
	(3)	0.23	0.87	1.00	0.73	0.82	0.59	0.76	0.71	0.64	0.31	0.27
		[0.70]	[3.52]	[3.82]	[2.76]	[3.01]	[3.37]	[4.43]	[4.22]	[3.78]	[1.98]	[1.74]
	(4)	0.55	0.81	0.86	0.80	0.81	0.27	0.36	0.34	0.29	-0.01	-0.03
		[1.89]	[3.66]	[3.64]	[3.20]	[3.18]	[1.82]	[2.50]	[2.36]	[2.01]	[-0.04]	[-0.22]
	(5)	0.57	0.63	0.88	0.71	0.82	0.25	0.25	0.38	0.36	0.33	0.31
		[2.21]	[3.40]	[4.60]	[3.75]	[3.20]	[1.47]	[1.44]	[2.80]	[2.57]	[2.33]	[2.18]

Panel B: Portfolio average number of firms and market capitalization

		DSC Quintiles Average n					DSC Quintiles Average market capitalization ($\$10^6$)				
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
Size quintiles	(1)	337	335	342	346	347	32	28	34	43	52
	(2)	101	101	101	101	101	66	71	73	71	73
	(3)	71	71	72	71	72	123	130	129	128	130
	(4)	62	62	62	62	62	276	298	288	281	285
	(5)	59	59	58	59	59	2197	2210	3043	2576	2040

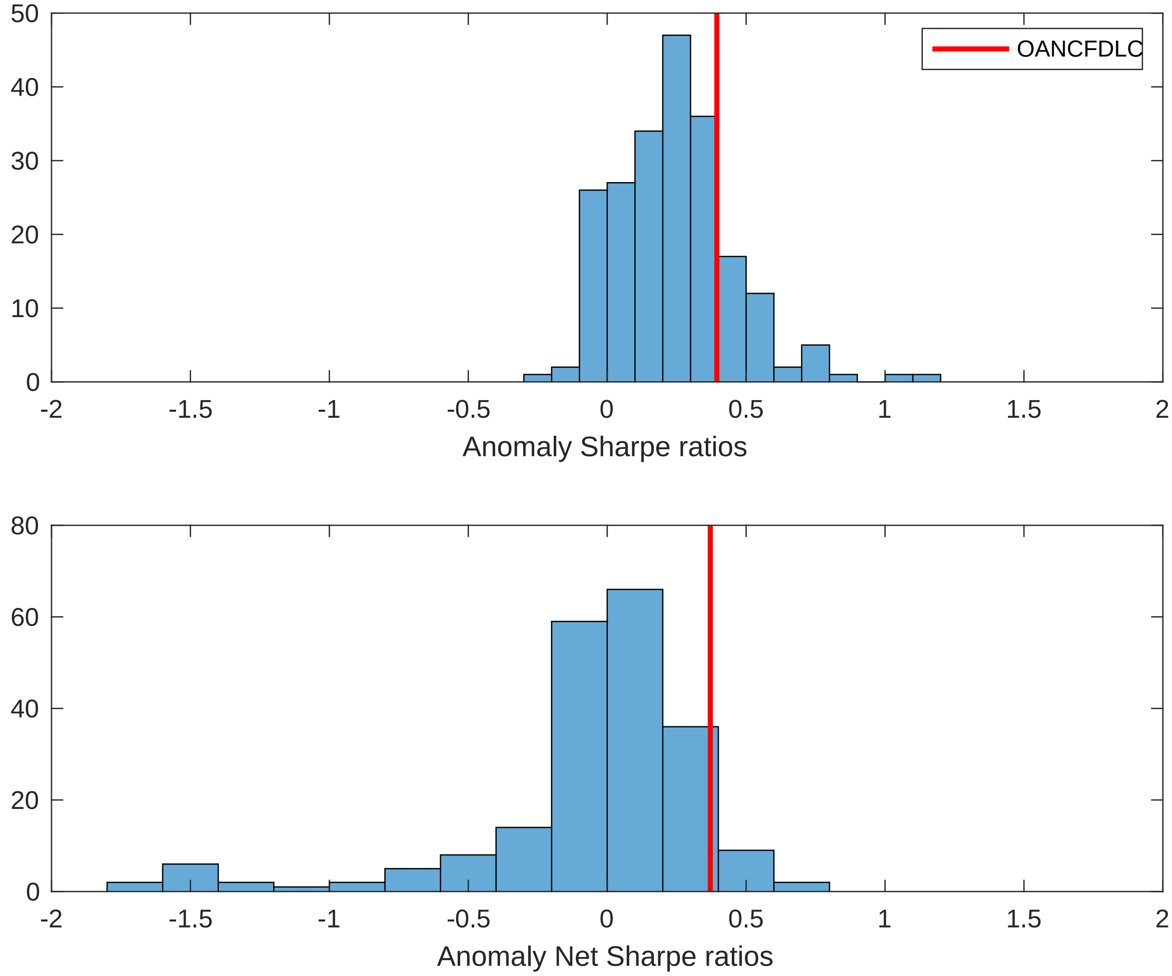


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the DSC with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

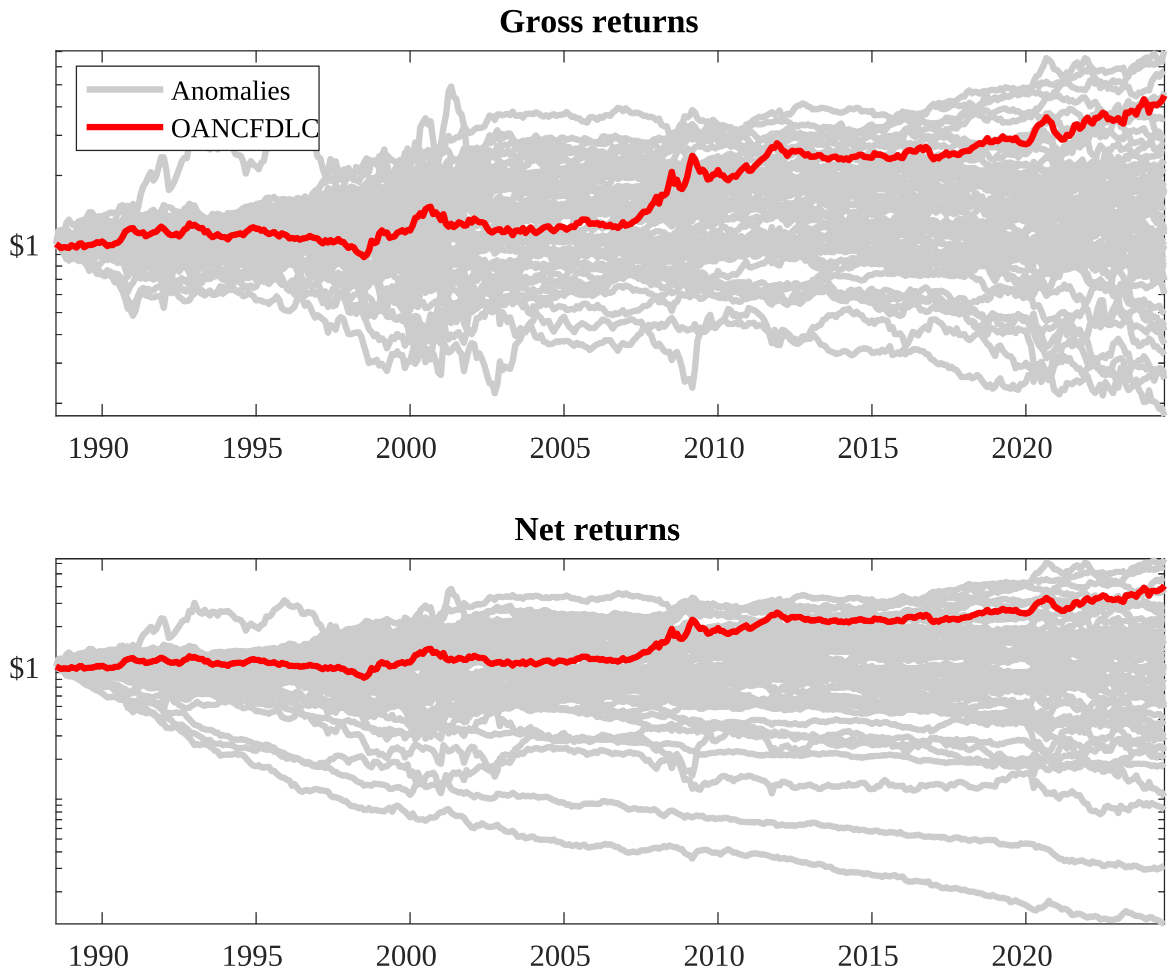
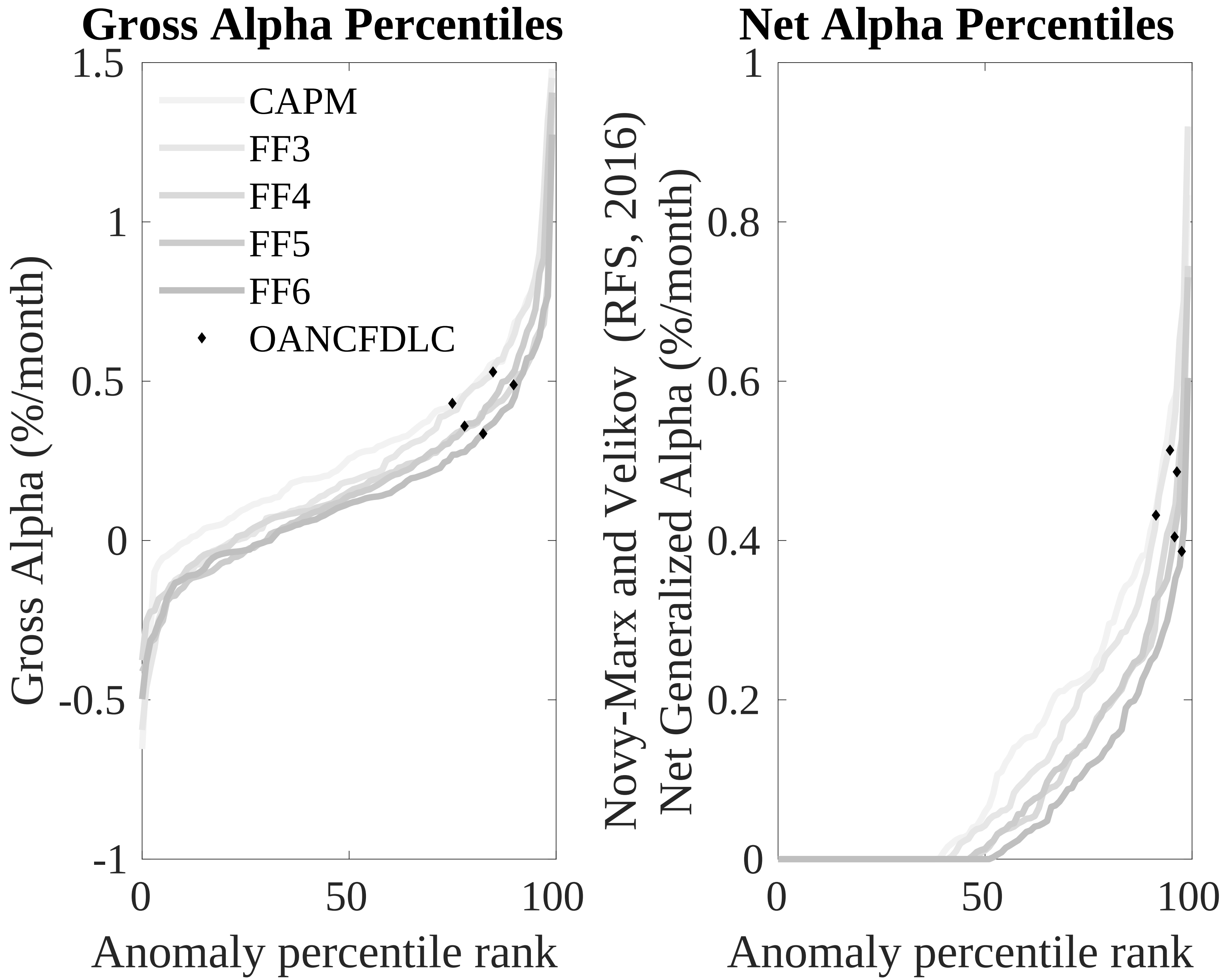


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the DSC trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Gross Alpha (%/month)
1.5
1
0.5
0
-0.5
-1
CAPM
FF3
FF4
FF5
FF6
OANCFDLC
0
50
100
Anomaly percentile rank
Net Alpha Percentiles
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
1
0.8
0.6
0.4
0.2
0
0
50
100
Anomaly percentile rank

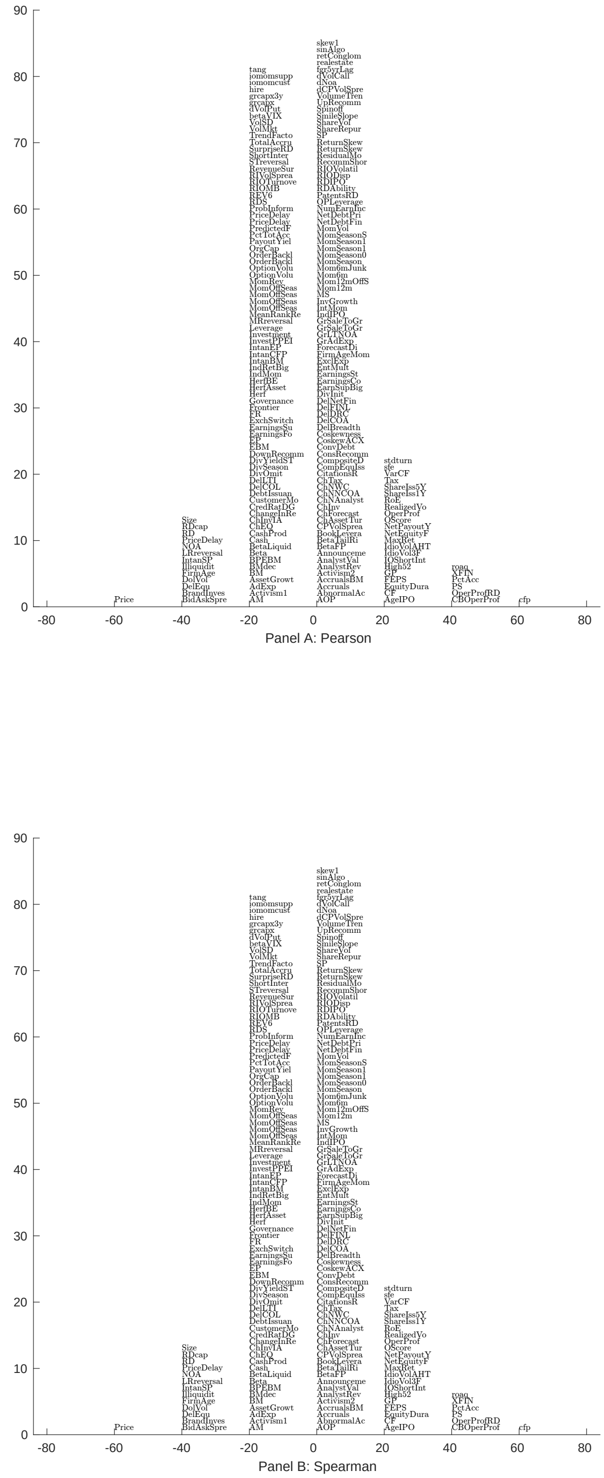


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 209 filtered anomaly signals with DSC. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

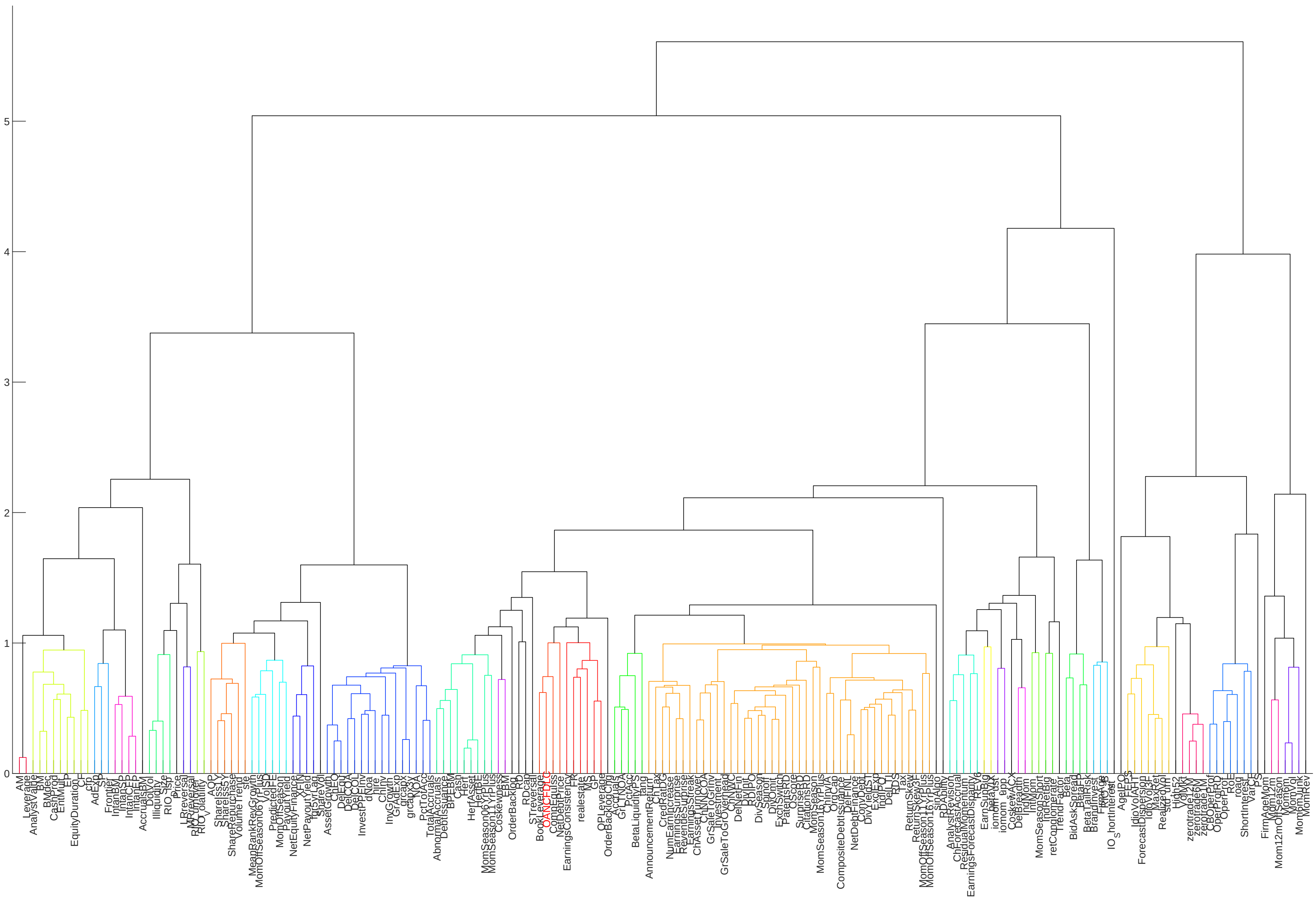


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

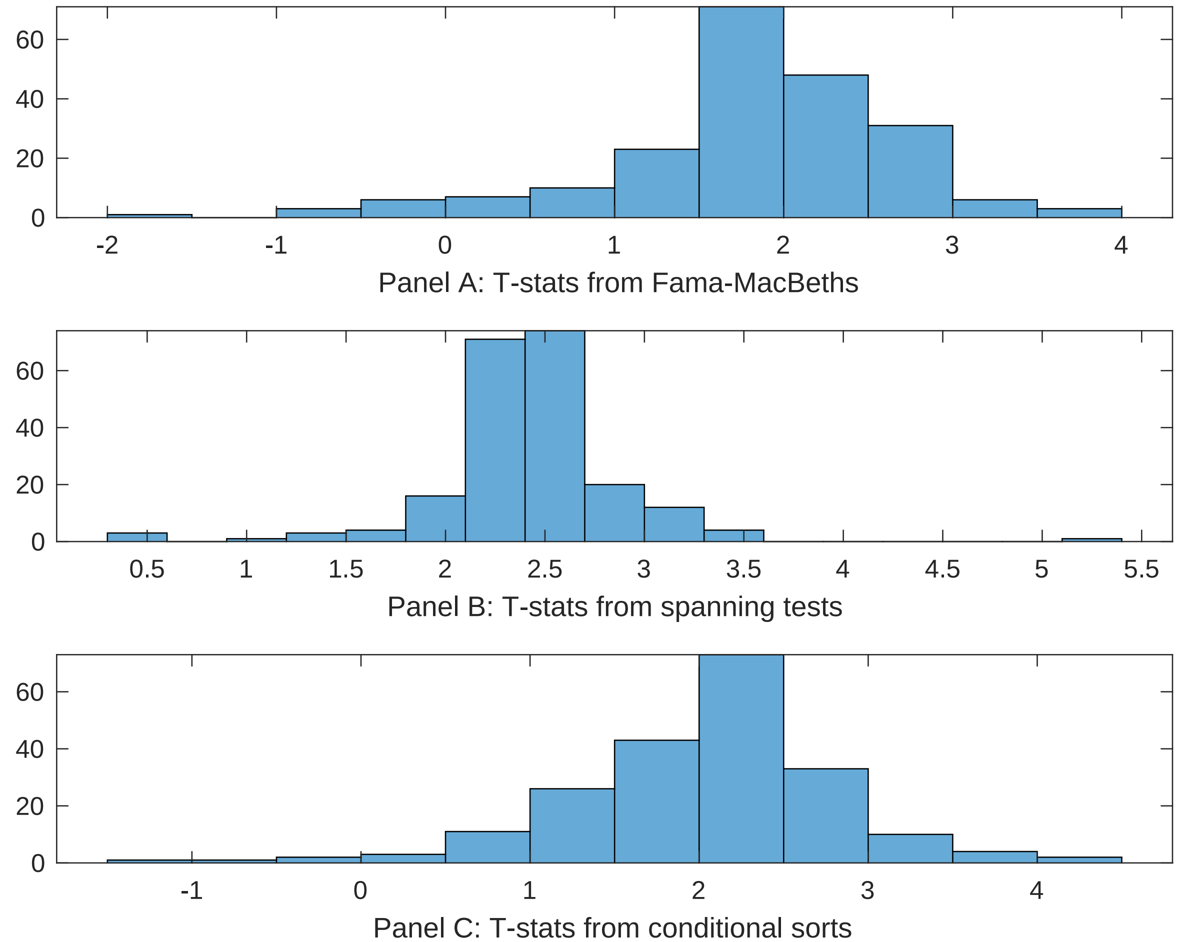


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of DSC conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DSC} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DSC} DSC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DSC,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on DSC. Stocks are finally grouped into five DSC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DSC trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on DSC. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{DSC} DSC_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Cash-based operating profitability, Operating Cash flows to price, Return on assets (qtrly), Net Operating Assets, Operating profitability RD adjusted, Accruals. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

Intercept	0.90	0.95	0.11	0.14	0.85	1.00	0.15
	[2.83]	[3.26]	[3.96]	[4.58]	[2.54]	[3.61]	[4.85]
DSC	0.24	0.71	-0.18	0.13	0.20	0.10	-0.46
	[0.05]	[1.42]	[-0.49]	[2.04]	[0.43]	[1.53]	[-1.35]
Anomaly 1	0.21						-0.70
	[4.70]						[-1.00]
Anomaly 2		0.10					0.46
		[3.88]					[1.63]
Anomaly 3			0.49				0.45
			[2.84]				[2.81]
Anomaly 4				0.53			0.87
				[3.76]			[7.79]
Anomaly 5					0.20		0.18
					[3.76]		[2.63]
Anomaly 6						0.13	0.17
						[3.09]	[2.79]
# months	427	427	427	432	427	432	427
$\bar{R}^2$(%)	1	1	1	0	1	0	0

Table 5: Spanning tests controlling for most closely related anomalies
This table presents spanning tests results of regressing returns to the DSC trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{DSC} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Cash-based operating profitability, Operating Cash flows to price, Return on assets (qtrly), Net Operating Assets, Operating profitability RD adjusted, Accruals. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198806 to 202406.

Intercept	0.25	0.32	0.30	0.59	0.29	0.24	0.32
	[1.86]	[2.33]	[2.25]	[5.10]	[2.12]	[2.08]	[3.19]
Anomaly 1	27.03						31.01
	[4.61]						[4.80]
Anomaly 2		11.55					4.32
		[2.17]					[1.10]
Anomaly 3			26.44				15.54
			[4.47]				[3.33]
Anomaly 4				-61.30			-49.69
				[-12.59]			[-10.80]
Anomaly 5					12.76		-4.57
					[2.28]		[-0.75]
Anomaly 6						57.51	37.07
						[13.39]	[9.10]
mkt	-2.31	-4.71	-0.79	-1.71	-2.51	-3.43	2.37
	[-0.69]	[-1.40]	[-0.23]	[-0.59]	[-0.72]	[-1.21]	[0.93]
smb	7.60	-0.66	6.37	-15.44	4.49	8.70	2.02
	[1.53]	[-0.14]	[1.30]	[-3.57]	[0.89]	[2.12]	[0.51]
hml	-55.57	-74.09	-57.44	-44.90	-60.90	-45.27	-23.10
	[-8.93]	[-10.66]	[-9.40]	[-8.45]	[-9.62]	[-8.72]	[-4.06]
rmw	12.78	22.58	2.39	21.79	16.97	40.31	3.66
	[1.91]	[3.55]	[0.30]	[4.17]	[2.30]	[7.72]	[0.56]
cma	13.42	14.79	17.09	23.71	17.93	-3.24	1.36
	[1.58]	[1.69]	[2.03]	[3.23]	[2.09]	[-0.44]	[0.21]
umd	2.03	9.50	-2.48	8.11	2.63	0.98	1.25
	[0.68]	[2.48]	[-0.75]	[3.15]	[0.85]	[0.39]	[0.42]
# months	428	428	428	432	428	432	428
$\bar{R}^2(\%)$	36	33	36	51	33	53	66

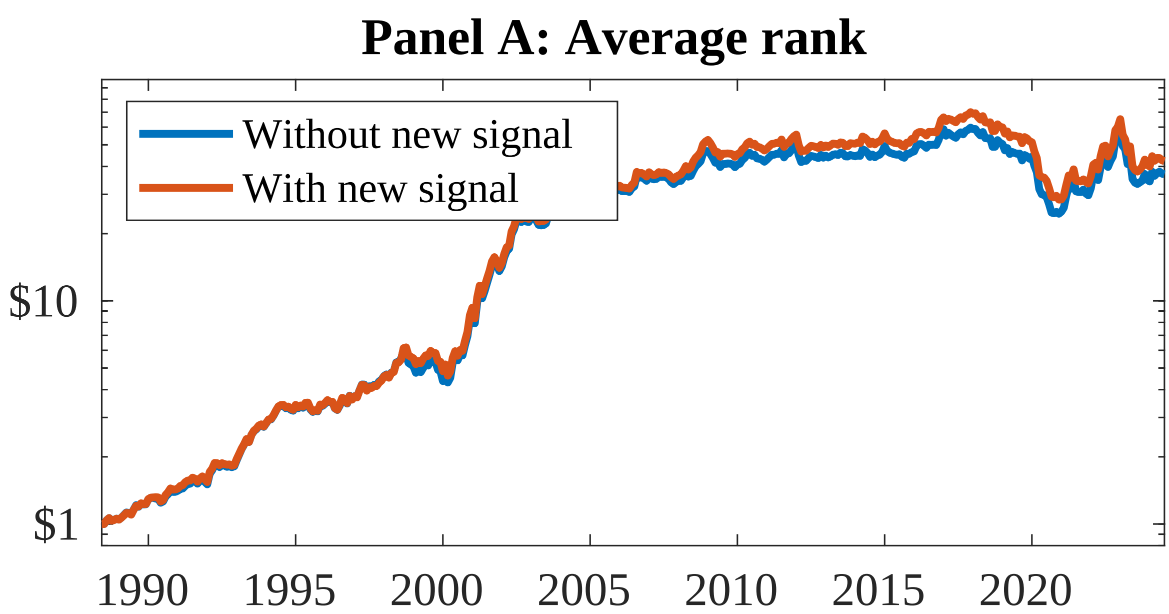


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 158 anomalies. The red solid lines indicate combination trading strategies that utilize the 158 anomalies as well as DSC. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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